Sum and Difference Formulas

Exact values with sum and difference formulas

1 Use a sum or difference formula to find the exact value of $\cos 75°$.

2 Find the exact value of $\sin \frac{\pi}{12}$.

3 Find the exact value of $\tan 105°$.

Working with given values

4 Angles A and B are both in Quadrant I with $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$. Find:

a $\sin(A+B)$

b $\cos(A+B)$

c the quadrant of $A+B$

5 Use a sum or difference formula to simplify:

a $\sin\left(x+\frac{\pi}{2}\right)$

b $\cos(x-\pi)$

6 Angles A and B are both in Quadrant II with $\sin A = \frac{8}{17}$ and $\sin B = \frac{3}{5}$. Find $\cos(A-B)$ exactly.

Verifying identities using sum and difference formulas

7 Verify the identity $\cos(x+y)+\cos(x-y)=2\cos x\cos y$.

8 Verify the identity $\sin(x+y)-\sin(x-y)=2\cos x\sin y$.

9 Show that $\tan\left(x+\frac{\pi}{4}\right)=\frac{1+\tan x}{1-\tan x}$ using the tangent sum formula $\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}$.

Applications

10 Two musical tones combine to form a sound wave modeled by $y=\sin\left(x+\frac{\pi}{6}\right)$. Expand this expression using the sum formula, and evaluate the coefficient of $\cos x$ exactly.

Additional practice

11 Find the exact value of $\cos\frac{\pi}{12}$.

12 Angles A and B satisfy $\tan A = \frac{3}{4}$ (QI) and $\tan B = \frac{5}{12}$ (QI). Find $\tan(A + B)$ using the tangent sum formula.

13 Simplify $\cos\left(\frac{\pi}{2} - x\right)\cos x + \sin\left(\frac{\pi}{2} - x\right)\sin x$ using the cosine difference formula in reverse.

A worked derivation

14 Starting from the cosine difference formula $\cos(A - B) = \cos A\cos B + \sin A\sin B$, derive the cosine sum formula $\cos(A + B)$ by substituting $B \to -B$.

15 Find the exact value of $\sin 15°\cos 15°$ using a sum/difference-related double angle shortcut (recall $\sin(2\theta) = 2\sin\theta\cos\theta$).

Sum and difference formulas with tables of values

16 Given $\sin A = \frac{7}{25}$ (QI) and $\cos B = \frac{4}{5}$ (QI), find $\sin(A - B)$ exactly.

17 Given $\tan A = 2$ (QI) and $\tan B = 3$ (QI), find $\tan(A - B)$ using the tangent difference formula.

Expanding a general sinusoidal expression

18 Expand $\cos\left(x - \frac{\pi}{3}\right)$ fully using the difference formula, leaving the answer in terms of $\sin x$ and $\cos x$.

A geometric interpretation

19 Explain, using the unit circle, why $\cos(A - B)$ (not $\cos A - \cos B$) correctly gives the cosine of the angle between two points at angles A and B on the unit circle.

Solving equations using sum and difference formulas

20 Solve $\sin\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$ for $x \in [0, 2\pi)$ by first expanding the left side.

21 Given that a signal is modeled by $y = \cos\left(3x - \frac{\pi}{4}\right)$, expand it using the difference formula and identify the coefficients of $\cos 3x$ and $\sin 3x$.

22 Two AC voltage signals are given by $V_1(t) = \sin(\omega t)$ and $V_2(t) = \sin\left(\omega t + \frac{2\pi}{3}\right)$ (phase-shifted by $120°$, common in three-phase power systems). Expand $V_2(t)$ fully using the sum formula, in terms of $\sin(\omega t)$ and $\cos(\omega t)$.